

Tutorial 9

9.2 A current source in a linear circuit has

$$i_s = 15 \cos(25\pi t + 25^\circ) \text{ A}$$

a) What is the amplitude of the current: 15 A

b) What is the angular frequency? $\omega = 25\pi = 78.54 \text{ rad/s}$

c) Find the frequency: $f = \frac{\omega}{2\pi} = 12.5 \text{ Hz}$

d) Calculate i_s at 2ms:

$$I_s = 15 \angle 25^\circ \text{ A}$$

$$\begin{aligned} I_s(2\text{ms}) &= 15 \cos(25\pi)(2 \times 10^{-3}) + 25^\circ \\ &= 15 \cos\left(\frac{1}{20}\pi + 25^\circ\right) \\ &= 15 \cos(25.15^\circ) \\ &= 13.58 \text{ A} \end{aligned}$$

9. For the following pairs of sinusoids, determine which one leads and by how much:

a) $v(t) = 10 \cos(4t - 60^\circ)$

$$\begin{aligned} i(t) &= 4 \sin(4t + 50^\circ) = 4 \cos(4t + 50^\circ - 90^\circ) \\ &= 4 \cos(4t - 40^\circ) \end{aligned}$$

$\therefore i(t)$ leads $v(t)$ by 20°

b) $v_1(t) = 4 \cos(377t + 10^\circ)$

$$v_2(t) = -20 \cos 377t = 20 \cos(377t + 180^\circ)$$

$\therefore v_2$ leads by 170°

$$\begin{aligned} \text{c) } x(t) &= 13 \cos 2t + 5 \sin 2t = 13 \cos 2t + 5 \cos(2t - 90^\circ) \\ &= 13 \angle 0^\circ + 5 \angle -90^\circ \end{aligned}$$

$$= 13 - j5 = 13.928 \angle -21.04^\circ$$

$$= 13.928 \cos(2t - 21.04^\circ)$$

$$y(t) = 15 \cos(2t - 11.8^\circ)$$

$$\therefore y(t) \text{ leads by } [-11.8 + 21.04 = 9.24^\circ]$$

9.8 Calculate these complex numbers and express your results in rectangular form:

$$\begin{aligned} \text{a) } \frac{60 \angle 45^\circ}{7.5 - j10} + j2 &= \frac{60 \angle 45^\circ}{12.5 \angle -53.13^\circ} + j2 \\ &= 4.8 \angle 98.13^\circ + j2 \\ &= -0.6788 + j4.751 + j2 \\ &= -0.6788 + j6.75 \end{aligned}$$

$$\begin{aligned} \text{b) } \frac{32 \angle -20^\circ}{(6 - j8)(4 + j2)} + \frac{20}{-10 + j24} \\ \therefore (6 - j8)(4 + j2) = 24 + j12 - j32 + 16 \\ = 40 - j20 = 20\sqrt{5} \angle -26.56^\circ \end{aligned}$$

$$\begin{aligned} \therefore \frac{32 \angle -20^\circ}{20\sqrt{5} \angle -26.56^\circ} + \frac{20(-10 - j24)}{(-10 + j24)(-10 - j24)} \\ = \frac{32 \angle -20^\circ}{20\sqrt{5} \angle -26.56^\circ} + \frac{-200 - j480}{100 + 576} \\ = 0.711 + j0.0817 - 0.2958 - j0.71 \\ = 0.415 - j0.6283 \end{aligned}$$

$$\begin{aligned}
 & c) 20 + (16 \angle -50^\circ)(5 + j12) \\
 & = 20 + (16 \angle -50^\circ)(13 \angle 67.38^\circ) \\
 & = 20 + 208 \angle 17.38^\circ \\
 & = 218.5 + j62.13
 \end{aligned}$$

9.13 Evaluate the following complex numbers:

$$\begin{aligned}
 & a) \frac{2 + j3}{1 - j6} + \frac{7 - j8}{-5 + j11} \\
 & = (-0.4324 + j0.4054) + (-0.8425 - j0.2534) \\
 & = -1.2749 + j0.1520
 \end{aligned}$$

$$\begin{aligned}
 & b) \frac{(5 \angle 10^\circ)(10 \angle -40^\circ)}{(4 \angle -80^\circ)(-6 \angle 50^\circ)} \\
 & = \frac{50 \angle -30^\circ}{24 \angle 150^\circ} = -2.083
 \end{aligned}$$

$$\begin{aligned}
 & c) \begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix} \\
 & = (2 + j3)(8 - j5) - (-4) = 35 + j14
 \end{aligned}$$

9.17 Two voltages v_1 and v_2 appear in series so that their sum is $V = v_1 + v_2$. If $v_1 = 10 \cos(50t - \pi/3)$ V and $v_2 = 12 \cos(50t + 30^\circ)$, find V :

$$\begin{aligned}
 V &= V_1 + V_2 = 10 \angle -60^\circ + 12 \angle 30^\circ \\
 &= 5 - j8.66 + 10.392 + j6 \\
 &= 15.62 \angle -9.805^\circ \\
 \therefore V &= 15.62 \cos(50t - 9.8^\circ)
 \end{aligned}$$

9.19 Using phasors find:

$$a) 3\cos(20t + 10^\circ) - 5\cos(20t - 30^\circ)$$

$$= 3\angle 10^\circ - 5\angle -30^\circ = 2.954 + j0.5209 - 4.33 + j2.5$$

$$= -1.376 + j3.021$$

$$= 3.32\angle 114.49^\circ$$

$$\therefore = 3.32\cos(20t + 114.49^\circ)$$

$$b) 40\sin 50t + 30\cos(50t - 45^\circ)$$

$$= 40\angle -90^\circ + 30\angle -45^\circ = -j40 + 21.21 - j21.21$$

$$= 21.21 - j61.21$$

$$= 64.78\angle -70.89^\circ$$

$$= 64.78\cos(50t - 70.89^\circ)$$

$$c) 20\sin 400t + 10\cos(400t + 60^\circ) - 5\sin(400t - 20^\circ)$$

$$\Rightarrow \sin\alpha = \cos(\alpha - 90^\circ)$$

$$= 20\angle -90^\circ + 10\angle 60^\circ - 5\angle -110^\circ = -j20 + 5 + j8.66 + 1.7101 + j4.69$$

$$= 6.7101 - j6.641$$

$$= 9.44\angle -44.7^\circ$$

$$= 9.44\cos(400t - 44.7^\circ)$$

9.20 A linear network has a current input:

$$7.5\cos(10t + 30^\circ)\text{A and a voltage output}$$

$$120\cos(10t + 75^\circ)\text{V. Determine impedance:}$$

$$I = 7.5\angle 30^\circ \quad V = 120\angle 75^\circ$$

$$\therefore Z = \frac{V}{I} = \frac{120\angle 75^\circ}{7.5\angle 30^\circ} = 16\angle 45^\circ$$